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 Studierichting: Informatica
 Oefeningenreeks 3G nr. 18.2

$$f(x) = \sqrt{\frac{x}{x+1}}$$

$$\text{dom } f = \mathbb{R} \setminus [-1, 0[$$

Nulpunten: $x = 0$

$$\text{Polen: } x + 1 = 0 \Leftrightarrow x = -1$$

VA:

$$\lim_{x \rightarrow -1+} \sqrt{\frac{x}{x+1}} = \frac{-1}{0^+} = -\infty$$

$$\lim_{x \rightarrow -1-} \sqrt{\frac{x}{x+1}} = \frac{-1}{0^-} = +\infty$$

$$\Rightarrow x = -1$$

HA:

$$\lim_{x \rightarrow \infty} \sqrt{\frac{x}{x+1}} = \frac{\infty}{\infty}$$

$$= \lim_{x \rightarrow \infty} \sqrt{\frac{\frac{x}{x}}{\frac{x}{x} + \frac{1}{x}}} = \lim_{x \rightarrow \infty} \sqrt{\frac{1}{1 + \frac{1}{x}}}$$

$$= \sqrt{\frac{1}{1+0}} = \sqrt{1} = 1$$

$$\Rightarrow y = 1.$$

x	$-\infty$	-1	0	$+\infty$
$f(x)$	$+$	$ $	0	$+$

De functie ligt naar $-\infty$ boven de HA en naar $+\infty$ onder de HA.

SA:

geen SA.

Tekenonderzoek:

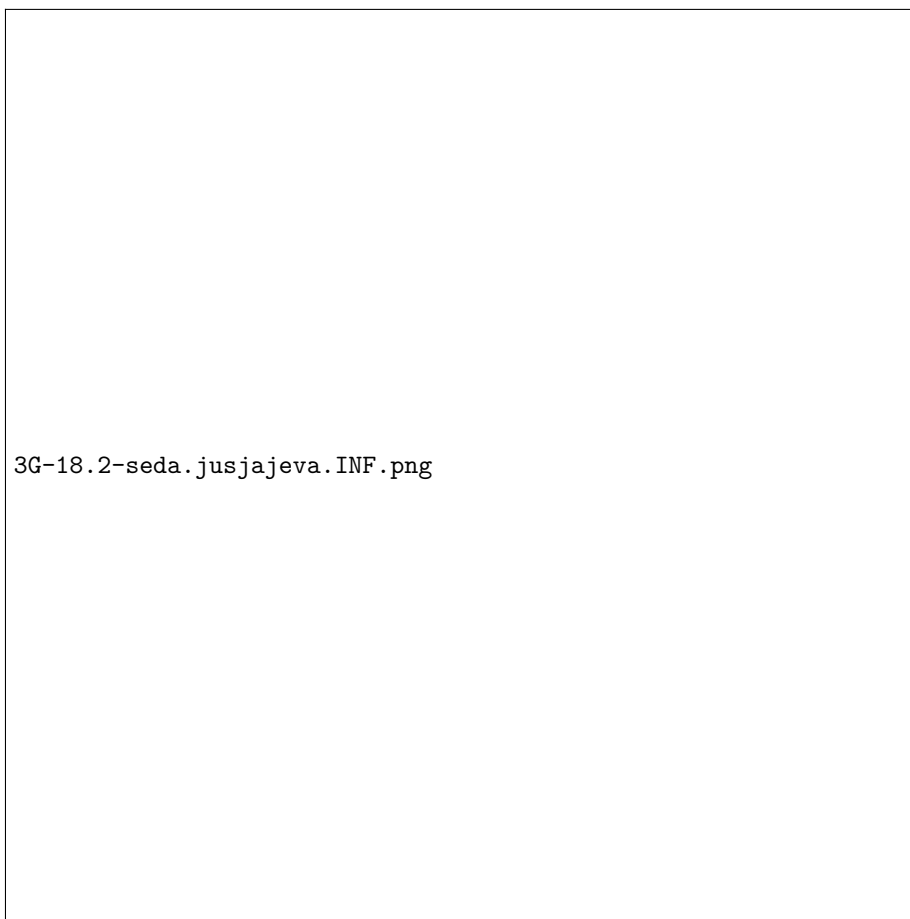
$$f'(x) = \frac{d}{dx} \left(\sqrt{\frac{x}{x+1}} \right)$$

$$= \frac{1}{2} \left(\frac{x}{x+1} \right)^{-\frac{1}{2}} \cdot \frac{d}{dx} \left(\frac{x}{x+1} \right)$$

$$\begin{aligned}
&= \frac{1}{2} \left(\frac{x}{x+1} \right)^{-\frac{1}{2}} \cdot \frac{1 \cdot (x+1) - x \cdot 1}{(x+1)^2} \\
&= \frac{1}{2} \left(\frac{x}{x+1} \right)^{-\frac{1}{2}} \cdot \frac{x+1-x}{(x+1)^2} \\
&= \frac{1}{2} \left(\frac{x}{x+1} \right)^{-\frac{1}{2}} \cdot \frac{1}{(x+1)^2} \\
&= \frac{1}{2} \sqrt{\frac{x+1}{x}} \cdot \frac{1}{(x+1)^2} \\
&= \frac{\sqrt{x+1}}{2\sqrt{x} \cdot (x+1)^2} \\
&= \frac{1}{2\sqrt{x} \cdot (x+1)^{\frac{3}{2}}} \\
f''(x) &= \frac{d}{dx} \left(\frac{1}{2\sqrt{x} \cdot (x+1)^{\frac{3}{2}}} \right) \\
&= \frac{1}{2} \cdot \frac{d}{dx} \left(\frac{1}{\sqrt{x} \cdot (x+1)^{\frac{3}{2}}} \right) \\
&= \frac{1}{2} \cdot \left(-\frac{1}{\left(\sqrt{x}(x+1)^{\frac{3}{2}} \right)^2} \cdot \frac{d}{dx} \left(\sqrt{x}(x+1)^{\frac{3}{2}} \right) \right) \\
&= \frac{1}{2} \cdot \left(-\frac{1}{x(x+1)^3} \cdot \left(\frac{d}{dx} \sqrt{x}(x+1)^{\frac{3}{2}} + \sqrt{x} \frac{d}{dx} (x+1)^{\frac{3}{2}} \right) \right) \\
&= \frac{1}{2} \cdot \left(-\frac{1}{x(x+1)^3} \cdot \left(\frac{(x+1)^{\frac{3}{2}}}{2\sqrt{x}} + \frac{3\sqrt{x}(x+1)^{\frac{1}{2}}}{2} \right) \right) \\
&= \frac{1}{2} \cdot \left(\left(-\frac{1}{x(x+1)^3} \cdot \frac{(x+1)^{\frac{3}{2}}}{2\sqrt{x}} \right) + \left(-\frac{1}{x(x+1)^3} \cdot \frac{3\sqrt{x}(x+1)^{\frac{1}{2}}}{2} \right) \right) \\
&= \frac{1}{2} \cdot \left(\left(-\frac{1}{2x^{\frac{3}{2}}(x+1)^{\frac{3}{2}}} \right) + \left(-\frac{3}{2x^{\frac{1}{2}}(x+1)^{\frac{5}{2}}} \right) \right) \\
&= \frac{1}{2} \cdot \left(\frac{-(x+1) - 3x}{2x^{\frac{3}{2}}(x+1)^{\frac{5}{2}}} \right) \\
&= -\frac{4x+1}{4x^{\frac{3}{2}}(x+1)^{\frac{5}{2}}}
\end{aligned}$$

x	-1	0
$f(x)$	$+$	0
$f'(x)$	$+$	$+$
$f''(x)$	$+$	$-$
$f(x)$	$\nearrow$ $\smile$	$\nearrow$ $\smile$

Grafiek:



References